

RES-5 framing correction: the one-loop is exact in the condensate; RES-5

TECT verification-first repository · claim B1-RH-ENUM

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1. Purpose and scope

While attempting the subtracted-susceptibility bound, a conflation in the preceding RES-5 notes was caught and is corrected here: the bare susceptibilities $\chi^{(k)} \sim \int G^k$ are NOT the loop expansion – they are the condensate-expansion coefficients of the EXACT one-loop free energy. This note disentangles the two expansions, locates RES-5 correctly (the loop expansion), and restates the residual. Scope: framing correction; anchor μ^2 , endpoint.

2. The one-loop is exact in the condensate

The Gaussian (one-loop) free energy in the condensate background is the log-det

$$F_{1\text{loop}}[\mathcal{P}] = \frac{1}{2} \text{Tr} \ln(D_0(I) + \lambda' \mathcal{P}^2),$$

which is SUMMED EXACTLY in the condensate; the selection margin $\Delta F^{(2)} = F_{1\text{loop}}[\mathcal{P}] - F_{1\text{loop}}[\mathcal{R}_H]$ is therefore exact at one-loop. Its condensate expansion $\frac{1}{2} \text{Tr} \sum_{k \geq 1} \frac{(-1)^{k-1}}{k} (\lambda' \mathcal{P}^2 D_0^{-1})^k$ has coefficients $\propto \int G^k$ and converges: the self-energy norm is $a_0/2 = \lambda' I / \hat{r} = 0.048$ on average and $\lambda' IN / \hat{r} = 0.574 < 1$ at the N^2 -enhanced constructive node,

$$\left\| \lambda' \langle \mathcal{P}^2 \rangle D_0^{-1} \right\|_{\text{avg}} = 0.048, \quad \left\| \lambda' \mathcal{P}^2 D_0^{-1} \right\|_{\text{node}} = 0.574 < 1,$$

so the expansion converges everywhere (even at the node). The growth of $\int G^{k+1} / \int G^k \rightarrow 1/\hat{r}$ is just the finite radius $\hat{r}$, NOT a divergence.

3. The bare-ratio "failure" was a conflation

The bare-route note read $\chi^{(3)}/\chi^{(2)} = 9.05$ (a condensate-expansion coefficient ratio of the EXACT one-loop) as an all-order RES-5 obstruction. That is a category error: the one-loop is exact, so its condensate-expansion coefficients carry no RES-5 information – one does not expand a quantity that is computed exactly. The NG-2026-06-09 entry is hereby annotated: the eliminated "bare-ratio route" was conflated; it does not constrain RES-5.

4. RES-5 is the loop expansion (the corrected residual)

Matched-order-to-exact (RES-5) is the LOOP expansion BEYOND the Gaussian: the two-loop (sunset), three-loop, etc. – the genuine fluctuation interactions. Of these:

- the **two-loop** difference is SC-SCOPE’s third cumulant, found thin ($\sim 4\%$, endpoint $\times 1.04$);
- the **higher loops** are the residual.

The common-mode cancellation (the strong common-sea loops cancel in the difference) and the a_0 suppression of the pattern-dependent part still apply – but now correctly to the LOOP-expansion difference, not the condensate expansion. The residual is therefore: bound the higher-loop DIFFERENCE $\Delta F^{(\ell\text{-loop})}$, $\ell \geq 3$, controlled by common-mode $+a_0$ – the SC-SCOPE programme extended to higher loops, via resummation of the strongly-fluctuating Brazovskii sea.

5. Status

T3 / FRAMING CORRECTION. The one-loop margin is exact; the bare-ratio route was a conflation (does not bear on RES-5); RES-5 is the loop expansion (SC-SCOPE two-loop done, higher loops the residual). No tier flip: B1-RH-ENUM T6 CONDITIONAL on {H-LAYER}. **Corrected residual:** the higher-loop difference bound (common-mode $+a_0$ controlled), a dedicated strong-coupling loop-resummation problem – not closeable by the elementary condensate-expansion or susceptibility-ratio attempts of this session.

6. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “*You flip-flopped – first the bare route failed, now it is irrelevant.*” CORRECT and that is the value: the bare-route note’s diagnosis (the $\chi^{(k)}$ are strong-coupling) was right, but its CONCLUSION (RES-5 obstruction) conflated the condensate expansion with the loop expansion. Catching and correcting this is the devil’s-advocate discipline working, not a defect hidden.
- (β) “*Is the one-loop really exact?*” YES: a Gaussian (one-loop) free energy is a determinant, computed exactly for a fixed background; the condensate-expansion convergence (peak $0.574 < 1$) confirms it is well-defined. The genuine non-perturbative content is the loop expansion.
- (γ) “*Does this change B1’s status?*” DISMISSED: no – it corrects the residual MAP (RES-5 = loop expansion), not any tier. B1 stays T6 on {H-LAYER}; no claim withdrawn.

Quantitative sanity check (mandatory): *convergence* – node self-energy $0.574 < 1$ (one-loop expansion converges); *average* – $a_0/2 = 0.048$ (weak on average); *growth* – $\int G^{k+1}/\int G^k \rightarrow 1/\hat{r} = 2.97$ (the finite radius, not a divergence); *consistency* – SC-SCOPE two-loop $\sim 4\%$ is the genuine RES-5 (loop) datum, far from the spurious bare 9.05.

7. Result footer

Result ID: B1-RH-ENUM / RES-5 one-loop vs loop-expansion disentangling
 Precise statement: $F_{1\text{loop}} = (1/2) \text{Tr} \ln(D_0 + \text{lam}' P^2)$ is EXACT in the condensate
 (expansion converges: avg self-energy $a_0/2 = 0.048$, peak
 node $\text{lam}' \text{IN}/\text{rhat} = 0.574 < 1$). The bare $\chi^{\sim}(k) \sim \int G^{\sim k}$ are

its condensate-expansion coefficients, NOT the loop expansion -- the bare-ratio "failure" ($\chi_3/\chi_2=9.05$) was a conflation. RES-5 = the LOOP expansion (2-loop+); SC-SCOPE third cumulant = the two-loop ($\sim 4\%$); the residual is the higher-loop difference (common-mode + a_0).

Scope: Framing correction; anchor μ^2 , endpoint.

Dependencies: res5-susceptibility-ratio-bareroute (corrected here); SC-SCOPE (two-loop); Math424-AddA.

Evidence grade: ANALYTIC + EXECUTED (res5_oneloop_disentangle.py 3/3).

Reproduction command: python codes/vacuum/res5_oneloop_disentangle.py

Expected output: selfE_avg=0.048, peak node 0.574<1; 3/3 PASS.

Falsification gate: A node self-energy ≥ 1 (one-loop expansion would diverge); or a demonstration that the loop expansion (not the condensate expansion) reduces to the bare χ^k ratio.

Tier before / after: No tier flip. B1 T6 on {H-LAYER}. RES-5 residual MAP corrected: loop expansion (higher loops), not the bare condensate-expansion ratio. NG-2026-06-09 annotated.

No-overclaim statement: This corrects a framing conflation; it does NOT close RES-5 and does NOT change B1's tier. The corrected residual (higher-loop difference, common-mode + a_0) is a dedicated strong-coupling loop-resummation problem.

Next required action: Treat the higher-loop difference directly (the SC-SCOPE two-loop programme extended): bound $\Delta F^{(l\text{-loop})}$, $l \geq 3$, via common-mode cancellation + a_0 suppression of the pattern-dependent part; resum the Brazovskii-sea loops.